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Algebra Applied Problems

An hour is 3,600 seconds. At 75 kilometers per hour, the car would take 3,600 seconds to travel 75 kilometers, so it would take $\frac{3600}{75} = 48$ seconds to travel 1 kilometer. But we're told that the car actually takes 2 seconds longer than that, or 50 seconds, to travel 1 kilometer. Therefore, in 1 hour, or 3,600 seconds, it travels $\frac{3600}{50} \times 1$ kilometer/hour = 72 kilometers/hour.

The correct answer is B.

•

If the speed of x meters per second is equivalent to the speed of y kilometers per hour, what is y in terms of x? (1 kilometer = 1,000 meters)

A. $\frac{5x}{18}$

B. $\frac{6x}{5}$

C. $\frac{18x}{5}$

D. $60x$

E. $3,600,000x$

Algebra Measurement Conversion

We are given that y kilometers per hour is the same speed as x meters per second. By converting from meters to kilometers using 1 km = 1,000 m and seconds to hours using 1 hr = 3,600 sec, we can obtain y in terms of x.

$$y\frac{\text{km}}{\text{hr}} = x\frac{\text{m}}{\text{sec}}$$

$$y\frac{\text{km}}{\text{hr}} = x\frac{\text{m}}{\text{sec}} \times \frac{1\text{ km}}{1{,}000\text{ m}} \times \frac{3{,}600\text{ sec}}{1\text{ hr}}$$

$$y\frac{\text{km}}{\text{hr}} = \left(\frac{3{,}600}{1{,}000}\right)x\frac{\text{km}}{\text{hr}}$$

$$y\frac{\text{km}}{\text{hr}} = \frac{18x}{5}\frac{\text{km}}{\text{hr}}$$

From the last equality above it follows that $y = \frac{18x}{5}$.

The correct answer is C.

•

The function f is defined by $f\left(x\right) = -\frac{1}{x}$ for all nonzero numbers x. If $f\left(a\right) = -\frac{1}{2}$ and $f\left(ab\right) = \frac{1}{6}$, then $b =$

A. 3

B. $\frac{1}{3}$

C. $-\frac{1}{x}$

D. –3

E. –12

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E. -12

Algebra Functions

From $f\left(x\right) = -\frac{1}{x}$ and $f\left(a\right) = -\frac{1}{2}$ we have $-\frac{1}{a} = -\frac{1}{2}$, or $a = 2$. From $f\left(x\right) = -\frac{1}{x}$ and $f\left(ab\right) = \frac{1}{6}$ we have $-\frac{1}{ab} = \frac{1}{6}$, or $ab = -6$. Since $a = 2$, it follows from $ab = -6$ that $2b = -6$, or $b = -3$.

The correct answer is D.

•

If $5^x - 5^{x-3} = (124)(5^y)$, what is y in terms of x?

A. x

B. $x - 6$

C. $x - 3$

D. $2x + 3$

E. $2x + 6$

Algebra Equations; Factoring

A strategy that is useful in rewriting sums and/or differences of exponential terms having the same base is to use the term with the least exponent as a common factor. For example, in $5^x - 5^{x-3}$, factor out 5^{x-3} since $x - 3$ is less than x. This gives $5^x - 5^{x-3} = 5^{x-3}(5^3 - 1)$. Therefore, the given equation is equivalent to $5^{x-3}(5^3 - 1) = (124)(5^y)$, or $5^{x-3}(124) = (124)(5^y)$. Dividing both sides of the last equation by 124 gives $5^{x-3} = 5^y$, and hence $y = x - 3$.

The correct answer is C.

•

The total of Company C's assets in 1994 was 300 percent greater than the total in 1993, which in turn was 400 percent greater than the total in 1992. If the total of Company C's assets in 1992 was N dollars, which of the following represents the total of Company C's assets, in dollars, in 1994?

A. $7N$

B. $8N$

C. $9N$

D. $12N$

E. $20N$

Algebra Percents

The total in 1993 was 400% greater than N; that is, it was $5N$. The total in 1994 was 300% greater than the total in 1993; that is, it was 4 times the total in 1993. Thus, the total in 1994 was $4(5N) = 20N$.

The correct answer is E.

•

If $\frac{x}{|y|} = -1$, which of the following must be true?

A. $x = -y$

B. $x = y$

C. $x = y^2$

D. $x^2 = y^2$

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D. $x^2 = y^2$

E. $x^3 = y^3$

Algebra Equations

When $y \neq 0$, the given equation is equivalent to $x = (-1)|y|$. Since $|y|$ is equal to y or $-y$, depending on whether $y > 0$ or $y < 0$, and $y^2 = (-y)^2$, it follows that $|y|^2 = y^2$. Therefore, squaring both sides of $x = (-1)|y|$ gives $x^2 = (-1)^2|y|^2 = y^2$.

Alternatively, if $x = -1$, then for both $y = 1$ and $y = -1$ we have $\frac{x}{|y|} = -1$. The table shows that among the answer choices, only $x^2 = y^2$ is true for both of these choices of x and y.

	$x = -y$	$x = y$	$x = y^2$	$x^2 = y^2$	$x^3 = y^3$
$x = -1$ and $y = 1$	true	false	false	**true**	false
$x = -1$ and $y = -1$	false	true	false	**true**	true

The correct answer is D.

A	1	$2\sqrt{3}$
B	$\sqrt{6}$	C
$\sqrt{3}$	6	D

•

The table above is a 3 by 3 grid of 9 numbers, 5 of which are given. The 4 numbers that are not given are denoted by A, B, C, and D. The product of the numbers in each row and in each column is the same for all rows and columns. What is the value of the product $ABCD$?

A. 36

B. 72

C. $36\sqrt{2}$

D. $36\sqrt{3}$

E. $72\sqrt{6}$

Algebra Simultaneous Equations; Computation with Radical Expressions

For each of the 3 rows and 3 columns, let x be the common product of the numbers in that row or column. The table below shows the products associated with the 3 rows and 3 columns. For example, the product of the numbers in the right-most column is $(2\sqrt{3})(C)(D)$, and so we have $2\sqrt{3}CD = x$.

row products	column products
$2\sqrt{3}A = x$	$\sqrt{3}AB = x$
$\sqrt{6}BC = x$	$6\sqrt{6} = x$
$6\sqrt{3}D = x$	$2\sqrt{3}CD = x$

Multiplying together the 3 left-hand sides and the 3 right-hand sides of the 3 column product equations gives $(\sqrt{3}AB)(6\sqrt{6})(2\sqrt{3}CD) = x^3$, or $36\sqrt{6}ABCD = x^3$. Since $x = 6\sqrt{6}$ (the middle column product), it follows that

$36\sqrt{6}ABCD = (6\sqrt{6})^3$. Therefore,

$$ABCD = \frac{(6\sqrt{6})\ (6\sqrt{6})\ (6\sqrt{6})}{(36\sqrt{6})} = 36.$$

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$$(36\sqrt{6})$$

Alternatively, begin with

$(\sqrt{3}AB)(x)(2\sqrt{3}CD) = x^3$ instead of

$(\sqrt{3}AB)(6\sqrt{6})(2\sqrt{3}CD) = x^3$.

Now divide both sides by x and use

$(\sqrt{3})(2\sqrt{3}) = 6$ to get $6ABCD = (6\sqrt{6})^2$, or $ABCD = 36$.

The correct answer is A.

•

In May, Mrs. Lee's earnings were 60 percent of the Lee family's total income. In June, Mrs. Lee earned 20 percent more than in May. If the rest of the family's income was the same both months, then in June, Mrs. Lee's earnings were approximately what percent of the Lee family's total income?

A. 64%

B. 68%

C. 72%

D. 76%

E. 80%

Algebra Percents

Let x be the Lee family's total income in May. In May, Mrs. Lee earned $0.6x$, so the rest of the family's income was $(1-0.6x) = 0.4x$. In June, the rest of the family still just earned $0.4x$, but Mrs. Lee earned 20 percent more than in May; that is, she earned $1.2(0.6x) = 0.72x$. So in June, Mrs. Lee earned $\frac{0.72}{1.12} = \frac{72}{112} = \frac{9}{14} \approx 64\%$ of the family's income.

The correct answer is A.

•

John would have reduced the time it took him to drive from his home to a certain store by $\frac{1}{3}$ if he had increased his average speed by 15 miles per hour. What was John's actual average speed, in miles per hour, when he drove from his home to the store?

A. 25

B. 30

C. 40

D. 45

E. 50

Algebra Applied Problems

Let R be John's actual average speed (in miles per hour) and let T be John's actual travel time (in hours). Then it follows from ***distance = rate × time*** that the distance (in miles) from John's home to the store is RT. Since the same distance can be traveled when R is replaced by $R + 15$ and T is replaced by $\frac{2}{3}T$, we have

$RT = \left(R + 15\right)\left(\frac{2}{3}T\right)$. Now solve this eequation for R.

$$RT = (R + 15)\left(\frac{2}{3}T\right) \text{ from above}$$

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$R = \left(RT + 15\right)\left(\frac{2}{3}\right)$	divide both sides by T
$R = \frac{2}{3}R + 10$	expand right side
$\frac{1}{3}R = 10$	subtract $\frac{2}{3}R$ from both sides
$R = 30$	multiply both sides by 3

Therefore, John's actual average speed was 30 miles per hour.

The correct answer is B.

•

Which of the following inequalities has a solution set that, when graphed on the number line, is a single line segment of finite length?

A. $x^4 \geq 1$

B. $x^3 \leq 27$

C. $x^2 \geq 16$

D. $2 \leq |x| \leq 5$

E. $2 \leq 3x + 4 \leq 6$

Algebra Number Line

For option A, $x^4 \geq 1$, the solution set includes every $x \geq 1$ and also every $x \leq -1$. On the number line this is graphed as two line segments, one extending infinitely leftward from −1 and the other infinitely rightward from 1.

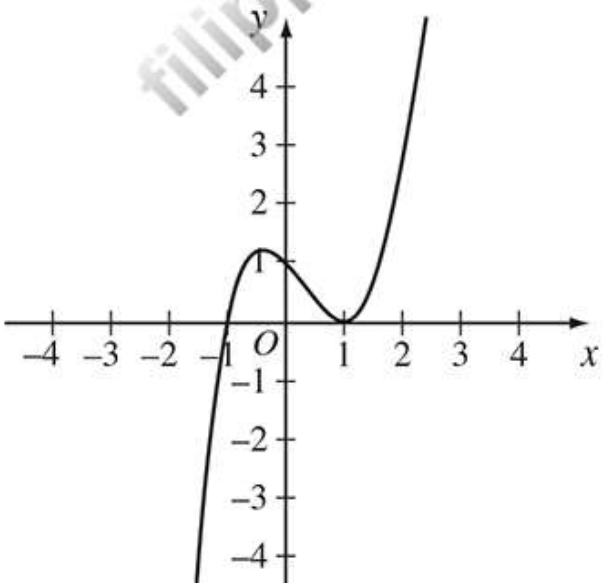
For option B, $x^3 \leq 27$, the solution set includes every $x \leq 3$. This is graphed as a line segment extending infinitely leftward from 3.

For option C, $x^2 \geq 16$, the solution set includes every $x \geq 4$ and also every $x \leq -4$. As with option A, this is graphed as two line segments extending infinitely in opposite directions.

For option D, $2 \leq |x| \leq 5$, the solution set includes every x from 2 through 5 and also every x from −2 through −5. This is graphed as two separate finite line segments.

For option E, $2 \leq 3x + 4 \leq 6$, the solution set includes only every x from $-\frac{2}{3}$ through $\frac{2}{3}$. This is graphed as a single line segment of finite length.

The correct answer is E.



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•

The figure shows the graph of $y = \left(x+1\right)(x-1)^2$ in the xy-plane. At how many points does the graph of $y = \left(x+1\right)(x-1)^2 + 2$ intercept the x-axis?

A. None

B. One

C. Two

D. Three

E. Four

Algebra Applied Problems

The graph of $y = \left(x+1\right)(x-1)^2 + 2$ is the graph of $y = \left(x+1\right)(x-1)^2$ shifted upward by 2. So, looking at the given graph of $y = \left(x+1\right)(x-1)^2$, we can see the graph of $y = \left(x+1\right)(x-1)^2 + 2$ intercepts the x-axis at only one point, left of the y-axis.

The correct answer is B.

Questions 177 to 272 — Difficulty: Hard

177. The cost to rent a small bus for a trip is x dollars, which is to be shared equally among the people taking the trip. If 10 people take the trip rather than 16, how many more dollars, in terms of x, will it cost per person?

A. $\frac{x}{6}$

B. $\frac{x}{10}$

C. $\frac{x}{16}$

D. $\frac{3x}{40}$

E. $\frac{3x}{80}$

Algebra Applied Problems

If 16 people take the trip, the cost per person would be $\frac{x}{16}$ dollars. If 10 people take the trip, the cost would be $\frac{x}{10}$ dollars. (Note that the lowest common multiple of 10 and 16 is 80.) Thus, if 10 people take the trip, the increase in dollars per person would be

$$\frac{x}{10} - \frac{x}{16} = \frac{8x}{80} - \frac{5x}{80} = \frac{3x}{80}.$$

The correct answer is E.

178. Clarissa will create her summer reading list by randomly choosing 4 books from the 10 books approved for summer reading. She will list the books in the order in which they are chosen. How many different lists are possible?

A. 6

B. 40

C. 210

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D. 5,040

E. 151,200

Arithmetic Elementary Combinatorics

Any of the 10 books can be listed first. Any of the 9 books remaining after the first book is listed can be listed second. Any of the 8 books remaining after the first and second books are listed can be listed third. Any of the 7 books remaining after the first, second, and third books are listed can be listed fourth. By the multiplication principle, there are (10)(9)(8)(7) = 5,040 different lists possible.

The correct answer is D.

179. If n is a positive integer and the product of all the integers from 1 to n, inclusive, is divisible by 990, what is the least possible value of n?

A. 8

B. 9

C. 10

D. 11

E. 12

Arithmetic Properties of Numbers

For convenience, let N represent the product of all integers from 1 through n. Then, since N is divisible by 990, every prime factor of 990 must also be a factor of N. The prime factorization of 990 is $2 \times 3^2 \times 5 \times 11$, and therefore, 11 must be a factor of N. Then the least possible value of N with factors of 2, 5, 3^2, and 11 is $1 \times 2 \times 3 \times \ldots \times 11$, and the least possible value of n is 11.

The correct answer is D.

180. The probability that event M will NOT occur is 0.8 and the probability that event R will NOT occur is 0.6. If events M and R CANNOT both occur, which of the following is the probability that either event M or event R will occur?

A. $\frac{1}{5}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $\frac{4}{5}$

E. $\frac{12}{25}$

Arithmetic Probability

Let $P(M)$ be the probability that event M will occur, let $P(R)$ be the probability that event R will occur, and let $P(M \text{ and } R)$ be the probability that events M and R both occur. Then the probability that either event M or event R will occur is $P(M) + P(R) - P(M \text{ and } R)$. From the given information, it follows that $P(M) = 1.0 - 0.8 = 0.2$, $P(R) = 1.0 - 0.6 = 0.4$, and $P(M \text{ and } R) = 0$. Therefore, the probability that either event M or event R will occur is $0.2 + 0.4 - 0 = 0.6 = \frac{3}{5}$.

The correct answer is C.

181. The total cost for Company X to produce a batch of tools is \$10,000 plus \$3 per tool. Each tool sells for \$8. The gross profit earned from producing and selling these tools is the total income from sales minus the total production cost. If a batch of 20,000 tools is produced and sold, then Company X's gross profit per tool is

A. \$2.00

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A. \$3.00

B. \$3.75

C. \$4.50

D. \$5.00

E. \$5.50

Arithmetic Applied Problems

The total cost to produce 20,000 tools is \$10,000 + \$3(20,000) = \$70,000. The revenue resulting from the sale of 20,000 tools is \$8(20,000) = \$160,000. The gross profit is \$160,000 – \$70,000 = \$90,000, and the gross profit per tool is $\frac{\$90,000}{20,000} = \$4.50$.

The correct answer is C.

182. If Q is an odd number and the median of Q consecutive integers is 120, what is the largest of these integers?

A. $\frac{Q-1}{2} + 120$

B. $\frac{Q}{2} + 119$

C. $\frac{Q}{2} + 120$

D. $\frac{Q+119}{2}$

E. $\frac{Q+120}{2}$

Arithmetic Statistics

For an odd number of data values, the median is the middle number. Thus, 120 is the middle number, and so half of the $Q-1$ remaining values are at most 120 and the other half of the $Q-1$ remaining values are at least 120. In particular, $\frac{Q-1}{2}$ data values lie to the right of 120 when the data values are listed in increasing order from left to right, and so the largest data value is $120 + \frac{Q-1}{2}$. Alternatively, it is evident that answer choice B, answer choice C, or answer choice E cannot be correct since these expressions do not have an integer value when Q is odd. For the list consisting of the single number 120 (i.e., if $Q = 1$), answer choice D fails because $\frac{Q+119}{2} = \frac{1+119}{2} = 60 \neq 120$ and answer choice A does not fail because $\frac{Q-1}{2} + 120 = \frac{1-1}{2} + 120 = 120$.

The correct answer is A.

183. If there are fewer than 8 zeros between the decimal point and the first nonzero digit in the decimal expansion of $\left(\frac{t}{1,000}\right)^4$, which of the following numbers could be the value of t?

I. 3

II. 5

III. 9

A. None

B. I only

C. II only

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C. II only

D. III only

E. II and III

Arithmetic Properties of Numbers; Decimals

Since $\left(\frac{t}{1{,}000}\right)^4 = \left(t \times \frac{1}{1{,}000}\right)^4 = t^4 \times \left(\frac{1}{1{,}000}\right)^4 = t^4 \times \frac{1}{1{,}000^4}$ and $(1{,}000)^4 = (10^3)^4 = 10^{12}$, it follows that $\left(\frac{t}{1{,}000}\right)^4 = t^4 \times \frac{1}{10^{12}} = t^4 \times 10^{-12}$. The following table illustrates the effect that multiplication by 10^{-1}, 10^{-2}, 10^{-3}, and 10^{-4} has on the placement of the decimal point of 52.7, a number chosen only for illustrative purposes.

multiplication by	resulting number	placement of decimal point
10^{-1}	5.27	1 place left of original
10^{-2}	0.527	2 places left of original
10^{-3}	0.0527	3 places left of original
10^{-4}	0.00527	4 places left of original

Therefore, the decimal point of $t^4 \times 10^{-12}$ is 12 positions to the left of the decimal point of t^4. Now consider the value of $t^4 \times 10^{-12}$ for the three given values of t.

t	t^4	$t^4 \times 10^{-12}$	# zeros after decimal point
3	81	0.000000000081	10
5	625	0.000000000625	9
9	6,561	0.000000006561	8

From the prior table it follows that NONE of these values of t is such that $t^4 \times 10^{-12}$ has fewer than 8 zeros between the decimal point and the first nonzero digit.

The correct answer is A.

184. A three-digit code for certain locks uses the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 according to the following constraints. The first digit cannot be 0 or 1, the second digit must be 0 or 1, and the second and third digits cannot both be 0 in the same code. How many different codes are possible?

A. 144

B. 152

C. 160

D. 168

E. 176

Arithmetic Elementary Combinatorics

Since the first digit cannot be 0 or 1, there are 8 digits possible for the first digit. Since the second digit must be 0 or 1, there are 2 digits possible for the second digit. If there were no other restrictions, all 10 digits would be possible for the third digit, making the total number of possible codes 8 × 2 × 10 = 160. But the additional restriction that the second and third digits cannot both be 0 in the same code eliminates the 8 codes 2-0-0, 3-0-0, 4-0-0, 5-0-0, 6-0-0, 7-0-0, 8-0-0, and 9-0-0. Therefore, there are 160 − 8 = 152 possible codes.

The correct answer is B.

185. Jackie has two solutions that are 2 percent sulfuric acid and 12 percent sulfuric acid by volume, respectively. If these solutions are mixed in appropriate quantities to produce 60 liters of a solution that is 5 percent sulfuric acid, approximately how many liters of the 2 percent solution will be required?

A. 18

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B. 20

C. 24

D. 36

E. 42

Algebra Simultaneous Equations

Let x represent the quantity of the 2% sulfuric acid solution in the mixture, from which it follows that the 2% sulfuric acid solution contributes $0.02x$ liters of sulfuric acid to the mixture. Let y represent the quantity of the 12% sulfuric acid solution in the mixture, from which it follows that the 12% sulfuric acid solution contributes $0.12y$ liters of sulfuric acid to the mixture. Since there are 60 liters of the mixture, $x + y = 60$. The quantity of sulfuric acid in the mixture, which is 5% sulfuric acid, is then (0.05)(60) = 3 liters. Therefore, $0.02x + 0.12y = 3$. Substituting $60 - x$ for y gives $0.02x + 0.12(60 - x) = 3$. Then,

$$\begin{aligned} 0.02x + 0.12(60 - x) &= 3 && \text{given} \\ 0.02x + 7.2 - 0.12x &= 3 && \text{use distributive property} \\ 7.2 - 0.1x &= 3 && \text{combine like terms} \\ -0.1x &= -4.2 && \text{subtract 7.2 from both sides} \\ x &= 42 && \text{divide both sides by } -0.1 \end{aligned}$$

The correct answer is E.

186. If Jake loses 8 pounds, he will weigh twice as much as his sister. Together they now weigh 278 pounds. What is Jake's present weight, in pounds?

A. 131

B. 135

C. 139

D. 147

E. 188

Algebra Systems of Equations

Let J represent Jake's weight and S represent his sister's weight. Then $J - 8 = 2S$ and $J + S = 278$. Solve the second equation for S and get $S = 278 - J$. Substituting the expression for S into the first equation gives

$$\begin{aligned} J - 8 &= 2(278 - J) \\ J - 8 &= 556 - 2J \\ J + 2J &= 556 + 8 \\ 3J &= 564 \\ J &= 188 \end{aligned}$$

The correct answer is E.

187. For each student in a certain class, a teacher adjusted the student's test score using the formula $y = 0.8x + 20$, where x is the student's original test score and y is the student's adjusted test score. If the standard deviation of the original test scores of the students in the class was 20, what was the standard deviation of the adjusted test scores of the students in the class?

A. 12

B. 16

C. 28

D. 36

E. 40

Arithmetic Statistics

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Arithmetic Statistics

The solution to this problem relies on the statistical properties summarized in the following table:

Data Set Values	Mean	Standard Deviation
x	μ	σ
$ax + b$	$a\mu + b$	$a\sigma$

The standard deviation of the original test scores was 20. Therefore, when the teacher multiplied each student's score by 0.8 and then added 20, the standard deviation of the set of adjusted scores is 0.8(20) = 16.

The correct answer is B.

188. Last year 26 members of a certain club traveled to England, 26 members traveled to France, and 32 members traveled to Italy. Last year no members of the club traveled to both England and France, 6 members traveled to both England and Italy, and 11 members traveled to both France and Italy. How many members of the club traveled to at least one of these three countries last year?

A. 52

B. 67

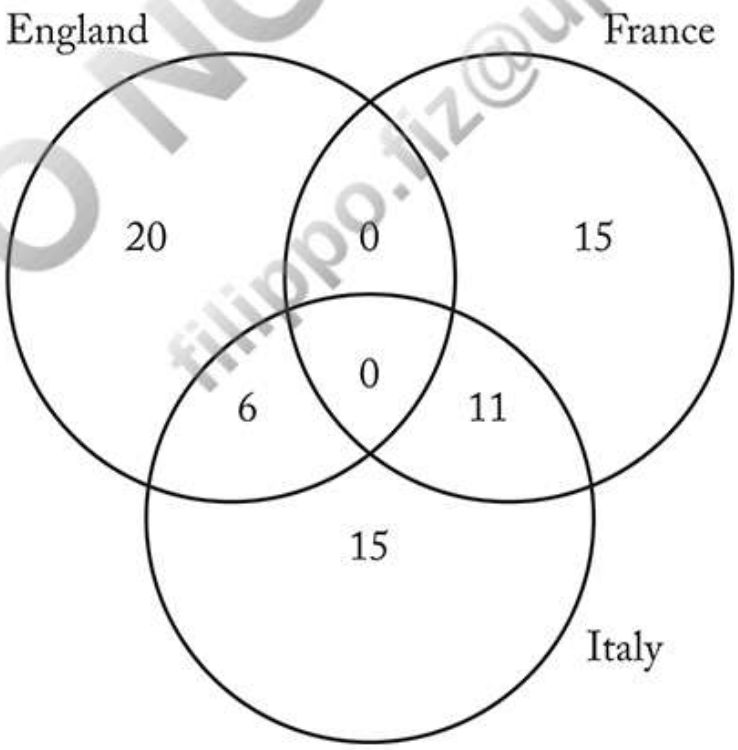
C. 71

D. 73

E. 79

Arithmetic Applied Problems

The numbers in the Venn diagram below represent the numbers of members of the club who traveled to the indicated countries, and these numbers can be determined as follows. Since no members traveled to both England and France, both regions that form the overlap of England and France are labeled with 0. It follows that none of the 6 members who traveled to both England and Italy traveled to France, and so the region corresponding to England and Italy only is labeled with 6. It also follows that none of the 11 members who traveled to both France and Italy traveled to England, and so the region corresponding to France and Italy only is labeled with 11. At this point it can be determined that 26 – 6 = 20 members traveled to England only, 26 – 11 = 15 members traveled to France only, and 32 – (6 + 11) = 15 members traveled to Italy only.



From the diagram it follows that 20 + 15 + 6 + 11 + 15 = 67 members traveled to at least one of these three countries.

The correct answer is B.

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The correct answer is B.

189. A store reported total sales of $385 million for February of this year. If the total sales for the same month last year was $320 million, approximately what was the percent increase in sales?

A. 2%

B. 17%

C. 20%

D. 65%

E. 83%

Arithmetic Percents

The percent increase in sales from last year to this year is 100 times the quotient of the difference in sales for the two years divided by the sales last year. Thus, the percent increase is

$$\begin{aligned}\frac{385-320}{320}\times 100 &= \frac{65}{320}\times 100\\ &= \frac{13}{64}\times 100\\ &\approx \frac{13}{65}\times 100\\ &= \frac{1}{5}\times 100\\ &= 20\%\end{aligned}$$

The correct answer is C.

190. When positive integer x is divided by positive integer y, the remainder is 9. If $\frac{x}{y} = 96.12$, what is the value of y?

A. 96

B. 75

C. 48

D. 25

E. 12

Arithmetic Properties of Numbers

The remainder is 9 when x is divided by y, so $x = yq + 9$ for some positive integer q. Dividing both sides by y gives $\frac{x}{y} = q + \frac{9}{y}$. But $\frac{x}{y} = 96.12 = 96 + 0.12$

Equating the two expressions for $\frac{x}{y}$ gives $q + \frac{9}{y} = 96 + 0.12$. Thus, $q = 96$ and $\frac{9}{y} = 0.12$.

$9 = 0.12y$

$y = \frac{9}{0.12}$

$y = 75$

The correct answer is B.

191. If $x(2x+1) = 0$ and $\left(x + \frac{1}{2}\right)\left(2x - 3\right) = 0$, then $x =$

A. –3

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B. $-\frac{1}{2}$

C. 0

D. $\frac{1}{2}$

E. $\frac{3}{2}$

Algebra Second-Degree Equations; Simultaneous Equations

Setting each factor equal to 0, it can be seen that the solution set to the first equation is $\left\{0, -\frac{1}{2}\right\}$ and the solution set to the second equation is $\left\{-\frac{1}{2}, \frac{3}{2}\right\}$. Therefore, $-\frac{1}{2}$ is the solution to both equations.

The correct answer is B.

192. A certain experimental mathematics program was tried out in 2 classes in each of 32 elementary schools and involved 37 teachers. Each of the classes had 1 teacher and each of the teachers taught at least 1, but not more than 3, of the classes. If the number of teachers who taught 3 classes is n, then the least and greatest possible values of n, respectively, are

A. 0 and 13

B. 0 and 14

C. 1 and 10

D. 1 and 9

E. 2 and 8

Algebra Simultaneous Equations; Inequalities

It is given that 2(32) = 64 classes are taught by 37 teachers. Let k, m, and n be the number of teachers who taught, respectively, 1, 2, and 3 of the classes. Then $k + m + n = 37$ and $k + 2m + 3n = 64$. Subtracting these two equations gives $m + 2n = 64 - 37 = 27$, or $2n = 27 - m$, and therefore $2n \leq 27$. Because n is an integer, it follows that $n \leq 13$ and answer choice B cannot be the answer. Since $n = 0$ is possible, which can be seen by using $m = 27$ and $k = 10$ (obtained by solving $2n = 27 - m$ with $n = 0$, then by solving $k + m + n = 37$ with $n = 0$ and $m = 27$), the answer must be answer choice A.

It is not necessary to ensure that $n = 13$ is possible to answer the question. However, it is not difficult to see that $k = 23$, $m = 1$, and $n = 13$ satisfy the given conditions.

The correct answer is A.

193. For the positive numbers n, $n + 1$, $n + 2$, $n + 4$, and $n + 8$, the mean is how much greater than the median?

A. 0

B. 1

C. $n + 1$

D. $n + 2$

E. $n + 3$

Algebra Statistics

Since the five positive numbers n, $n + 1$, $n + 2$, $n + 4$, and $n + 8$, are in ascending order, the median is the third number, which is $n + 2$. The mean of the five numbers is

$$\frac{n + (n + 1) + (n + 2) + (n + 4) + (n + 8)}{5}$$

$$5n + 15$$

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$$= \frac{5n + 15}{5}$$
$$= n + 3$$

Since $(n+3) - (n+2) = 1$, the mean is 1 greater than the median.

The correct answer is B.

194. The present ratio of students to teachers at a certain school is 30 to 1. If the student enrollment were to increase by 50 students and the number of teachers were to increase by 5, the ratio of students to teachers would then be 25 to 1. What is the present number of teachers?

A. 5

B. 8

C. 10

D. 12

E. 15

Algebra Applied Problems

After noting that $\frac{\text{number of students}}{\text{number of teachers}} = \frac{30}{1}$ implies that the number of students is 30 times the number of teachers, this problem can be solved with arithmetic, as shown in the table below, by guessing the present number of teachers, increasing the numbers of students and teachers as specified, then checking to see if the resulting ratio is equal to $\frac{25}{1}$.

Present Number of Teachers	Present Number of Students	Teachers Increased by 5	Students Increased by 50	Resulting Ratio	Equal to $\frac{25}{1}$?
5	150	10	200	$\frac{200}{10} = \frac{20}{1}$	no
8	240	13	290	$\frac{290}{13} = \frac{22\frac{4}{13}}{1}$	no
10	300	15	350	$\frac{350}{15} = \frac{23\frac{1}{3}}{1}$	no
12	360	17	410	$\frac{410}{17} = \frac{24\frac{2}{17}}{1}$	no
15	450	20	500	$\frac{500}{20} = \frac{25}{1}$	yes

Therefore, the present number of teachers is 15.

Alternatively, the problem can be solved using algebra.

The following table summarizes the given information, where s represents the present number of students and t represents the present number of teachers:

	Number of Students	Number of Teachers	$\frac{\text{Students}}{\text{Teachers}}$	Equation After Cross Multiplying
Present	s	t	$\frac{s}{t} = \frac{30}{1}$	$s = 30t$
After Increases	$s + 50$	$t + 5$	$\frac{s+50}{t+5} = \frac{25}{1}$	$s + 50 = 25(t+5)$

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